

DATA 23700

Project Deliverable 3: Project Defense

Defenses held during the last week of instruction. Slides due on Gradescope at the start of the group's assigned defense slot.

Each group defends their completed project in a private oral exam with the instructor. The group gives a short pitch for their project, and then the instructor asks questions about it. The defense is the group's opportunity to show that they understand the work they did and can account for the choices behind it. Slides accompany the pitch and are submitted as a separate deliverable.

Format

- **Length:** 20 minutes total, consisting of a 5-minute pitch followed by 15 minutes of questions from the instructor. The 5-minute limit on the pitch is strictly enforced.
- **Attendance:** all group members must attend the group's defense. Every member should expect to answer questions.
- **Scheduling:** defenses are held in private, one group at a time, during the last week of instruction. Groups claim a slot using a shared sign up spreadsheet on a first-come basis. The instructor may rebalance the schedule to keep the sessions even.

The pitch

The pitch is a compressed account of the project. In five minutes, it should cover:

- **Motivation:** the problem or question the project addresses, and why it's interesting.
- **Approach:** the dataset, analysis, and visualizations used, and any nontrivial modeling or data-wrangling choices.
- **Findings:** what the analysis revealed, illustrated by two or three of the group's most effective visualizations.
- **Implications:** what the group takes away from the analysis (e.g., conclusions, call to action, limitations of their approach, or next steps).

Groups are welcome to be creative about how they structure and sequence this material. The list above is a checklist of what to cover, not a required outline. The pitch should be empirically grounded in the analysis documented in the group's project repository.

The questions

The 15 minutes of questions are open-ended and follow from the group's own project. The instructor will ask about whatever the project raises: where the data came from and what it actually represents, the question the group posed and how they arrived at it, how the data was wrangled and modeled, why the group made the analysis and design choices they made rather than the alternatives, what conclusions the results do and do not support, and what the group would do differently.

No aspect of a group's project is off limits. This document is not a list of questions to prepare answers for, and groups should not treat it as one. Groups should expect to discuss their project in depth and without a script. The best preparation is to have done the work and to understand it.

Slides

Slides are the visual companion to the pitch and are graded as part of this deliverable. They should be:

- **Readable:** text and figures large enough to read comfortably on a shared screen.
- **Visually consistent:** coherent color palette, typography, and layout.
- **Focused:** minimal text, clear takeaways, one idea per slide when possible.
- **Anchored in the data:** include visualizations of the patterns and relationships in the chosen dataset that are key to the group's work.

Groups may use PowerPoint, Keynote, Google Slides, or any other tool of their choice, but must submit slides as a PDF to Gradescope by the start of the group's assigned defense slot.

Grading criteria

The first three criteria are scored for the group as a whole. The fourth is scored individually, so members of the same group may receive different scores on it.

- **Clarity of project narrative:** the group gives a concise and persuasive pitch for their project. It is clear why they did this work, how they did it, and what they learned.
- **Design of presentation materials:** slides and visualizations are well-designed, reflecting practices taught in this course.
- **Defensibility of analysis and design choices:** the group can defend their design and analysis choices in discussion with the instructor.
- **Individual ownership:** each group member receives an individual score reflecting the ownership they demonstrate over the project. This reflects whether a member can account substantively for the work they contributed and can engage with questions about the project as a whole.

The defense may *not* be resubmitted.

If a group member cannot attend

If a group member is unable to attend the group's scheduled defense (e.g., illness), the group should proceed without them and email the instructor in advance if possible. Absent members should contact the instructor to arrange an alternative demonstration of their contribution. The general late work exception policies in the syllabus apply.